Lab 05

★ **Implement SQL Operators: Demonstrate the use of different types of operators in SQL with examples, including Arithmetic Operators, Logical Operators, Comparison Operators.**

```
CREATE TABLE Product (
   ProductID INT,
   ProductName VARCHAR(50),
   Price DECIMAL(10,2),
   Quantity INT,
   Discount DECIMAL(10,2)
);

INSERT INTO Product VALUES
(1, 'Laptop', 50000, 2, 5000),
(2, 'Mobile', 20000, 3, 2000),
(3, 'Keyboard', 1500, 5, 100),
(4, 'Mouse', 800, 4, 50);
SELECT * FROM Product;
```

```
1   CREATE TABLE Product (
2       ProductID INT,
3       ProductName VARCHAR(50),
4       Price DECIMAL(10,2),
5       Quantity INT,
6       Discount DECIMAL(10,2)
7   );
8
9   INSERT INTO Product VALUES
10  (1, 'Laptop', 50000, 2, 5000),
11  (2, 'Mobile', 20000, 3, 2000),
12  (3, 'Keyboard', 1500, 5, 100),
13  (4, 'Mouse', 800, 4, 50);
14  SELECT * FROM Product;
15
```

Output:

```
+-----------+-------------+----------+----------+----------+
| ProductID | ProductName | Price    | Quantity | Discount |
+-----------+-------------+----------+----------+----------+
|         1 | Laptop      | 50000.00 |        2 |  5000.00 |
|         2 | Mobile      | 20000.00 |        3 |  2000.00 |
|         3 | Keyboard    |  1500.00 |        5 |   100.00 |
|         4 | Mouse       |   800.00 |        4 |    50.00 |
+-----------+-------------+----------+----------+----------+
```

```
SELECT 'ARITHMETIC OPERATIONS' AS Operations;

SELECT ProductName, Price * Quantity AS TotalAmount

FROM Product;

SELECT ProductName, Price / Quantity AS DividedPrice

FROM Product;

SELECT ProductName, Quantity %2 AS Remainder
```

FROM Product;

SELECT 'Comparison Operators' AS Operations;

SELECT * FROM Product

WHERE Price > 10000;

SELECT * FROM Product

WHERE Price < 10000;

SELECT * FROM Product

WHERE Price >= 20000;

SELECT 'Logical Operators' AS Operations;

SELECT * FROM Product

WHERE Price > 10000 AND Quantity >= 2;

SELECT * FROM Product

WHERE Price > 40000 OR Quantity >= 5;

SELECT * FROM Product

WHERE NOT Price > 10000;

```
16  SELECT 'ARITHMETIC OPERATIONS' AS Operations;
17  SELECT ProductName, Price * Quantity AS TotalAmount
18  FROM Product;
19  SELECT ProductName, Price / Quantity AS DividedPrice
20  FROM Product;
21  SELECT ProductName, Quantity % 2 AS Remainder
22  FROM Product;
23
24
25  SELECT 'Comparison Operators' AS Operations;
26  SELECT * FROM Product
27  WHERE Price > 10000;
28  SELECT * FROM Product
29  WHERE Price < 10000;
30  SELECT * FROM Product
31  WHERE Price >= 20000;
32
33  SELECT 'Logical Operators' AS Operations;
34  SELECT * FROM Product
35  WHERE Price > 10000 AND Quantity >= 2;
36  SELECT * FROM Product
37  WHERE Price > 40000 OR Quantity >= 5;
38  SELECT * FROM Product
39  WHERE NOT Price > 10000;
```

Operations
ARITHMETIC OPERATIONS

ProductName	TotalAmount
Laptop	100000.00
Mobile	60000.00
Keyboard	7500.00
Mouse	3200.00

ProductName	DividedPrice
Laptop	25000.000000
Mobile	6666.666667
Keyboard	300.000000
Mouse	200.000000

ProductName	Remainder
Laptop	0
Mobile	1
Keyboard	1
Mouse	0

Operations
Logical Operators

ProductID	ProductName	Price	Quantity	Discount
1	Laptop	50000.00	2	5000.00
2	Mobile	20000.00	3	2000.00

ProductID	ProductName	Price	Quantity	Discount
1	Laptop	50000.00	2	5000.00
3	Keyboard	1500.00	5	100.00

ProductID	ProductName	Price	Quantity	Discount
3	Keyboard	1500.00	5	100.00
4	Mouse	800.00	4	50.00

```
+----------------------+
| Operations           |
+----------------------+
| Comparison Operators |
+----------------------+
```

ProductID	ProductName	Price	Quantity	Discount
1	Laptop	50000.00	2	5000.00
2	Mobile	20000.00	3	2000.00

ProductID	ProductName	Price	Quantity	Discount
3	Keyboard	1500.00	5	100.00
4	Mouse	800.00	4	50.00

ProductID	ProductName	Price	Quantity	Discount
1	Laptop	50000.00	2	5000.00
2	Mobile	20000.00	3	2000.00